

Week 3 Plan Explanation – Risk Metrics & Portfolio Construction

Week Goal

Learn essential risk measures and implement basic portfolio optimization.

By the end of the week, the intern can measure portfolio risk, explain diversification, compare equal-weight, risk-parity, and optimized allocations, and identify where estimation error matters.

Task 1: Risk Metrics

This task implements the core measures needed to describe portfolio risk, reward, and drawdown behavior consistently.

Detailed Work

- Volatility: annualized standard deviation of returns
- Sharpe ratio: $(R_p - R_f) / \sigma_p$
- Maximum drawdown: largest peak-to-trough decline
- Drawdown duration: how long to recover

Concrete Output

- Implement `compute_risk_metrics(returns)` in `src/risk.py` returning a dictionary of all metrics
- Apply to the quintile portfolios from Week 2

Quality Check

Metric definitions, risk-free-rate treatment, frequency, and annualization must be consistent across every portfolio comparison.

Task 2: Correlation & Diversification

This task demonstrates how correlation, beta, and the number of holdings shape diversification at the portfolio level.

Detailed Work

- Correlation matrix of individual stock returns
- Diversification ratio: weighted average vol / portfolio vol
- Portfolio beta: $\text{cov}(R_p, R_m) / \text{var}(R_m)$

Concrete Output

- Compute and plot the correlation heatmap for 20 stocks
- Show how portfolio volatility decreases as you add more stocks (diversification curve)
- Compute beta of each quintile portfolio vs market

Quality Check

The correlation matrix and beta estimates must use aligned return dates, and the diversification curve must be generated with a clearly stated sampling rule.

Task 3: Equal-Weight vs Risk Parity

This task compares two transparent allocation rules and measures the trade-off between simplicity, concentration, and turnover.

Detailed Work

- Equal-weight portfolio: 1/N allocation
- Risk parity: allocate such that each asset contributes equally to portfolio risk
- Simple risk parity approximation: weight inversely proportional to volatility

Concrete Output

- Implement both allocation schemes
- Backtest both over the full sample period
- Compare: return, vol, Sharpe, max drawdown, turnover

Quality Check

Weights must sum to one at each rebalance, use only information then available, and report turnover alongside performance.

Task 4: Mean–Variance Optimization

This task introduces constrained mean–variance optimization while making estimation error a central part of the analysis.

Detailed Work

- Markowitz mean–variance framework
- Efficient frontier: plot risk vs return for various target returns
- Constraints: long–only, max weight per stock, sector limits
- Practical issues: estimation error in expected returns, covariance matrix

Concrete Output

- Implement simple mean–variance optimization using `scipy.optimize.minimize`
- Plot the efficient frontier with 50 stocks
- Compare the optimized portfolio vs equal–weight and risk parity

Quality Check

Constraints must be verified after optimization, the efficient frontier must be numerically feasible, and in–sample estimates must not be presented as forecasts.

Task 5: Review & Integration

This task consolidates the week's methods into reusable code and a decision–oriented comparison of the three portfolio approaches.

Detailed Work

- Discuss: why do simple methods (1/N, risk parity) often beat optimization in practice?
- Integrate risk functions into a reusable module

Concrete Output

- Create a chart pack (4–6 charts) comparing all three portfolio methods
- Write weekly memo summarizing findings
- Friday demo

Quality Check

The chart pack, module output, and memo tables must reconcile exactly and explain why simple methods may outperform optimized weights out of sample.

Week Deliverables

- `notebooks/week3/risk_portfolio_construction.ipynb`
- `src/risk.py` – risk metrics and portfolio optimization
- Chart pack: risk/return comparisons across methods
- Weekly memo (1 page)

Recommended Reading

- *Active Portfolio Management* (Grinold & Kahn), Chapter 2–3
- DeMiguel, Garlappi, & Uppal (2009), "Optimal versus naive diversification"
- Risk parity: Maillard, Roncalli, & Teiletche (2010)

Common Pitfalls

- Using in-sample expected returns for optimization (overfitting)
- Ignoring estimation error in covariance matrix
- Not annualizing metrics consistently (mixing daily and annual)

What Success Looks Like

The intern can measure portfolio risk, explain diversification, compare equal-weight, risk-parity, and optimized allocations, and identify where estimation error matters.